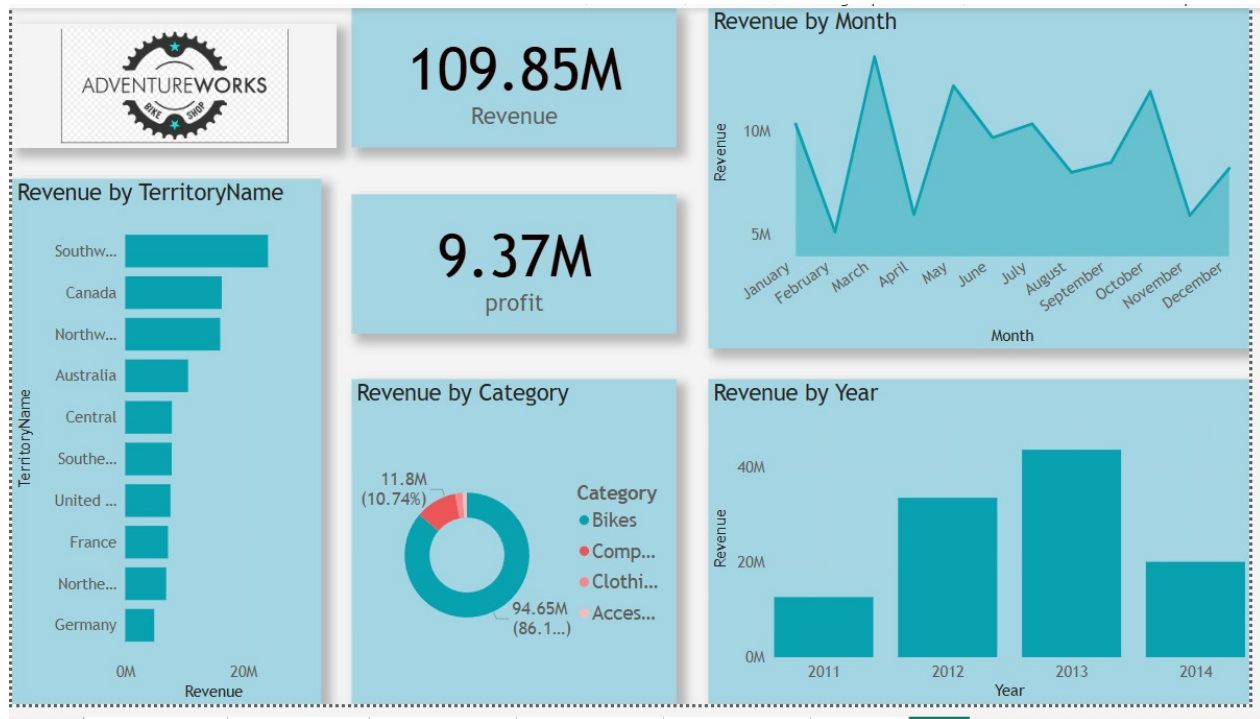


Sales Performance & Profitability Analysis

A Power BI Business Intelligence Case Study

Business objective

Turn strong top-line sales into more sustainable and profitable growth by identifying concentration risk, product portfolio inefficiency, profitability pressure, and gaps in operational measurement.



The business story behind the dashboard

The report reframes six Power BI dashboards as a decision-support case study rather than a chart-by-chart description.

\$109.85M Revenue	\$9.37M Profit	504 Products	121K Orders
19K Customers	\$905.45 Avg. Order Value	266 Products Sold	5 days Avg. Ship Duration

AdventureWorks shows strong commercial scale, but the dashboards reveal that the quality of growth deserves management attention. The central issue is not simply how to increase revenue; it is how to protect profit, reduce over-reliance on one category, improve portfolio productivity, and measure operations in a way that supports better decisions.

What the dashboards reveal

- Bikes contribute about 86% of displayed revenue.
- Only 266 of 504 products recorded sales in the dataset.
- Displayed total cost is \$100.47M against \$109.85M revenue.
- Revenue differs materially across sales territories.

Management questions created

- Which products and categories create the strongest profit, not just revenue?
- Why are 238 products not showing sales?
- Can non-bike categories grow through cross-sell and bundles?
- Which high-performing territory practices can be replicated?

Business problem statement

How can AdventureWorks convert strong sales activity into more resilient, profitable growth while improving product portfolio efficiency and management visibility?

Revenue is strong - but growth quality is uneven

Four analytical themes connect the dashboards into one management narrative.

Business issue	Evidence from dashboard	Why it matters
Category concentration	Bikes: about \$94.65M of \$109.85M revenue	A slowdown in one category could disproportionately affect total sales.
Portfolio productivity	266 sold products out of 504 total	Nearly half of the catalogue shows no recorded sales and needs classification.
Profitability pressure	\$9.37M profit vs. \$100.47M total cost	Management needs margin visibility below the company total.
Measurement quality	NULL categories; SUM-based shipping and size visuals	Some visuals can be misread if used directly for operational decisions.

Analytical approach

The case study distinguishes dashboard facts from interpretation. Where a visual does not support a causal or operational conclusion, the report flags the limitation and proposes a more decision-ready KPI.

Decision lens

Commercial performance

- Revenue trend and category mix
- Territory performance
- Customer and salesperson contribution
- Average order economics

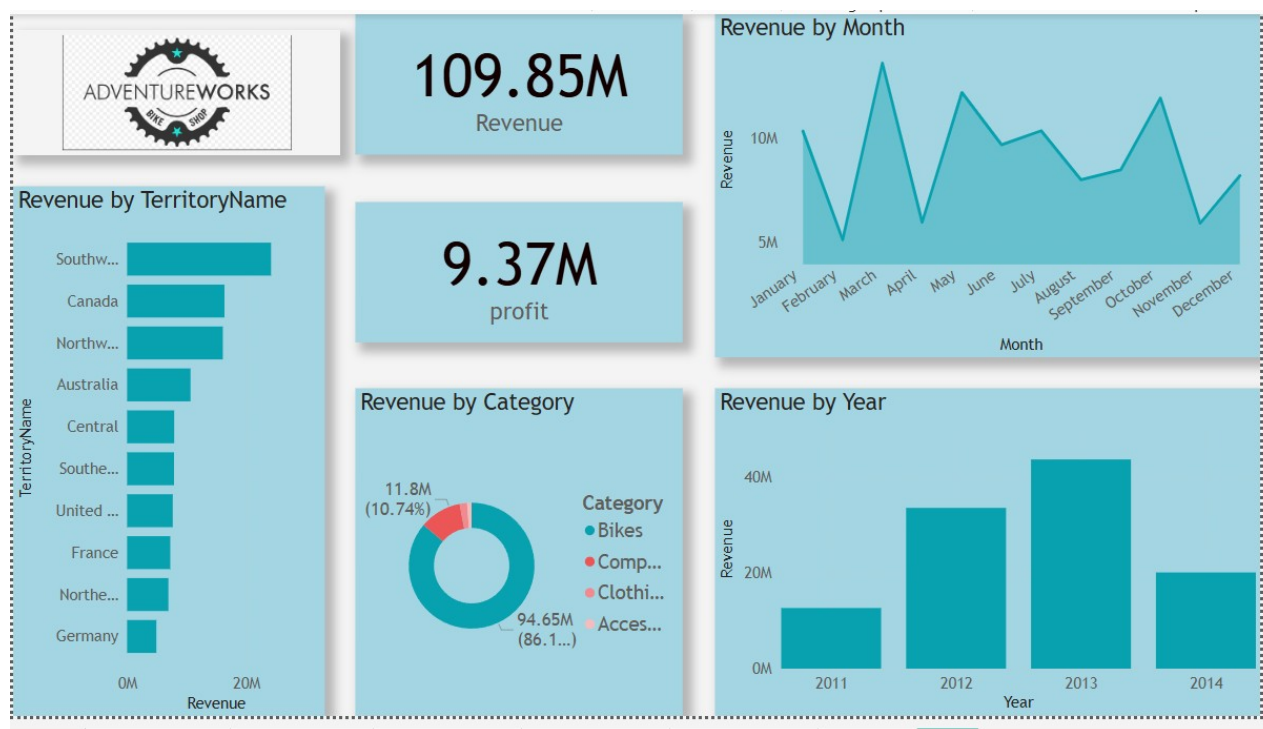
Efficiency and control

- Product portfolio utilisation
- Cost and profit logic
- Shipping performance measurement
- Data-quality completeness

Outcome: move from "what happened?" to "what should management investigate or change next?"

Strong revenue with significant category concentration

The overview dashboard is the starting point for assessing scale, trend and market dependence.



Evidence

- Total revenue displayed: \$109.85M.
- Profit displayed: \$9.37M.
- Bikes: about \$94.65M, or 86.1% of revenue.
- Components: about \$11.8M, or 10.7%.
- Southwest, Canada and Northwest are leading territories in the visual.

Business interpretation

- Bikes are the commercial engine, but they also create concentration risk.
- Non-bike categories represent the clearest diversification opportunity.
- Top territories should be benchmarked before investing in weaker markets.
- 2014 is lower than 2013 in the chart, but period completeness must be validated first.

Decision recommendation

Protect Bikes while developing complementary revenue from Components, Clothing and Accessories. Compare territory product mix, customer profile and order economics before choosing expansion actions.

Performance is concentrated by people and products - but context matters

The dashboard identifies visible leaders, while a management analysis should separate opportunity from structural advantage.



Evidence

- Roger Harui and Andrew Dixon appear among the top customers by revenue.
- Linda Mitchell and Jillian Carson lead the displayed salesperson revenue ranking.
- Mountain-200 variants dominate the top product revenue list.
- Salesperson activity rises through 2013 and is lower in the displayed 2014 period.

Better management questions

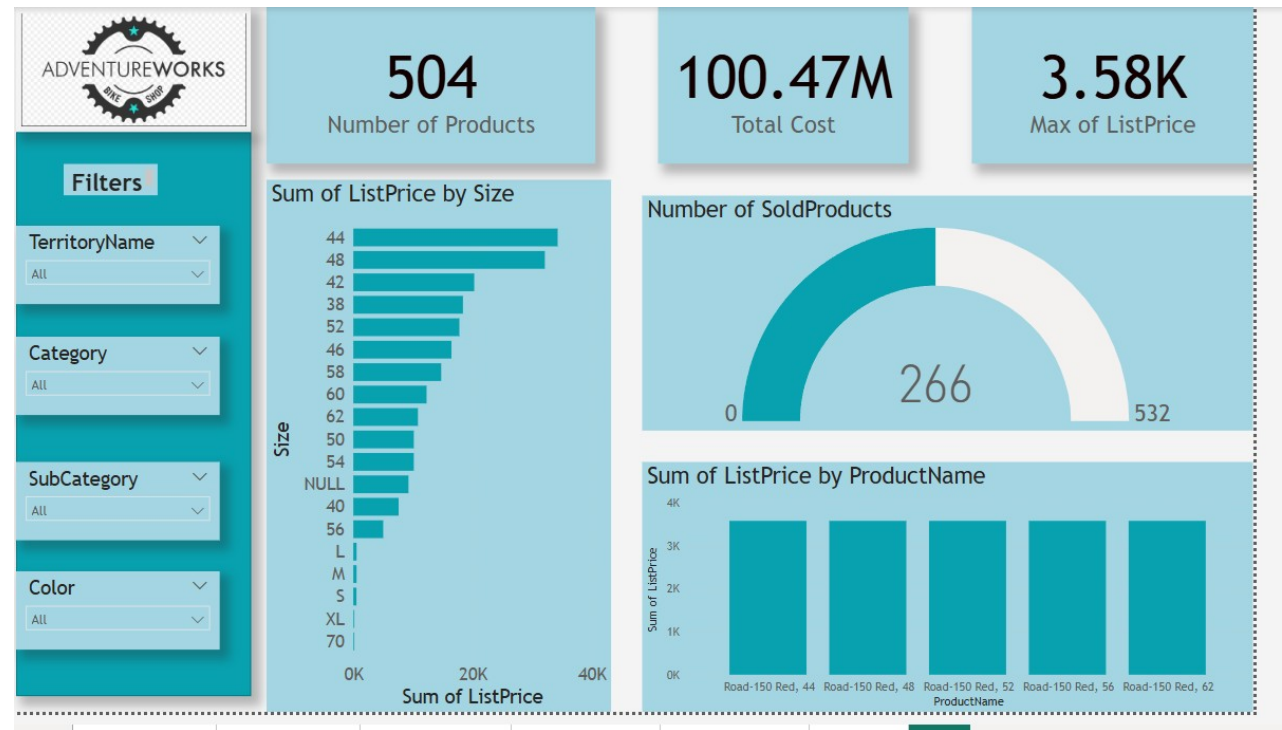
- Are top salespeople outperforming after adjusting for territory and customer mix?
- Which customers buy across multiple categories?
- Which Bike buyers have the highest cross-sell potential?
- Does performance come from customer volume, larger orders, or stronger retention?

Next KPI layer

Add Revenue per Active Customer, Profit per Salesperson, Repeat Purchase Rate, Average Order Value by Salesperson, and customer category penetration.

The clearest efficiency issue: only 52.8% of the catalogue recorded sales

The product dashboard exposes a portfolio utilisation problem and also highlights one visual that should not be used as a demand metric.



504	266	238	52.8%
Total products	With recorded sales	Without recorded sales	Catalogue utilisation

What is supported

- 266 / 504 = 52.8% of products recorded sales.
- 238 products therefore show no recorded sales.
- Road-150 Red variants appear among the highest list-price products.

What needs caution

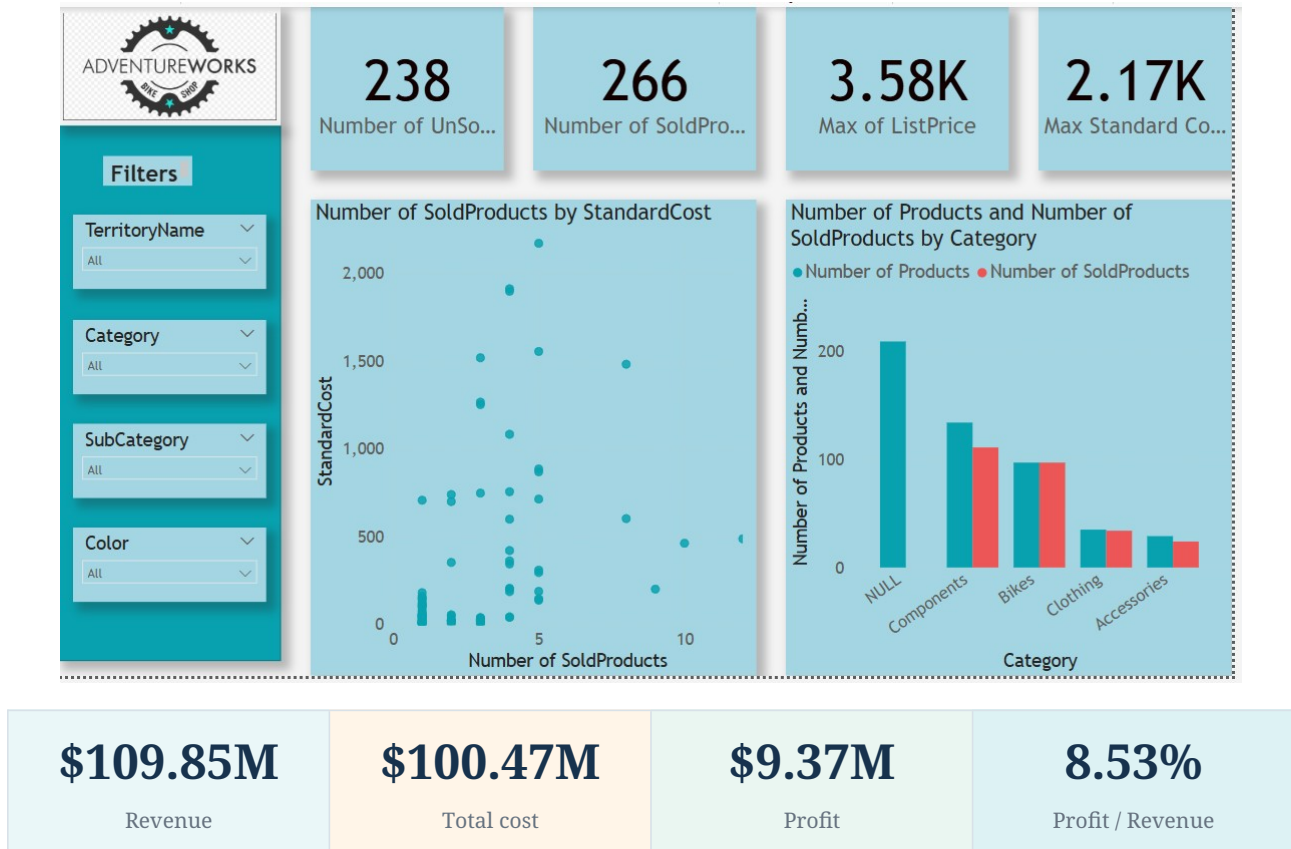
- Sum of ListPrice by Size is not a demand measure.
- A high value for size 44 or 48 can reflect expensive products, not more units sold.
- Unsold products may be new, seasonal, discontinued or affected by data quality.

Decision recommendation

Classify all 238 unsold products before rationalising the catalogue. Replace Sum of ListPrice by Size with Units Sold by Size and Revenue by Size for inventory decisions.

Validate KPI logic before using the headline percentage

The profitability dashboard is useful, but the displayed 9.33% requires interpretation.



Using the displayed totals, conventional profit margin is approximately $9.37 / 109.85 = 8.53\%$. The dashboard value 9.33 is approximately $9.37 / 100.47$, which is profit relative to cost rather than profit relative to revenue. For management reporting, these two measures should be labelled separately.

Recommended metric definitions

- Profit Margin = Profit / Revenue
- Return on Cost / Markup = Profit / Cost
- Gross Profit by Product and Category
- Margin % by Territory and Salesperson

Other issues visible

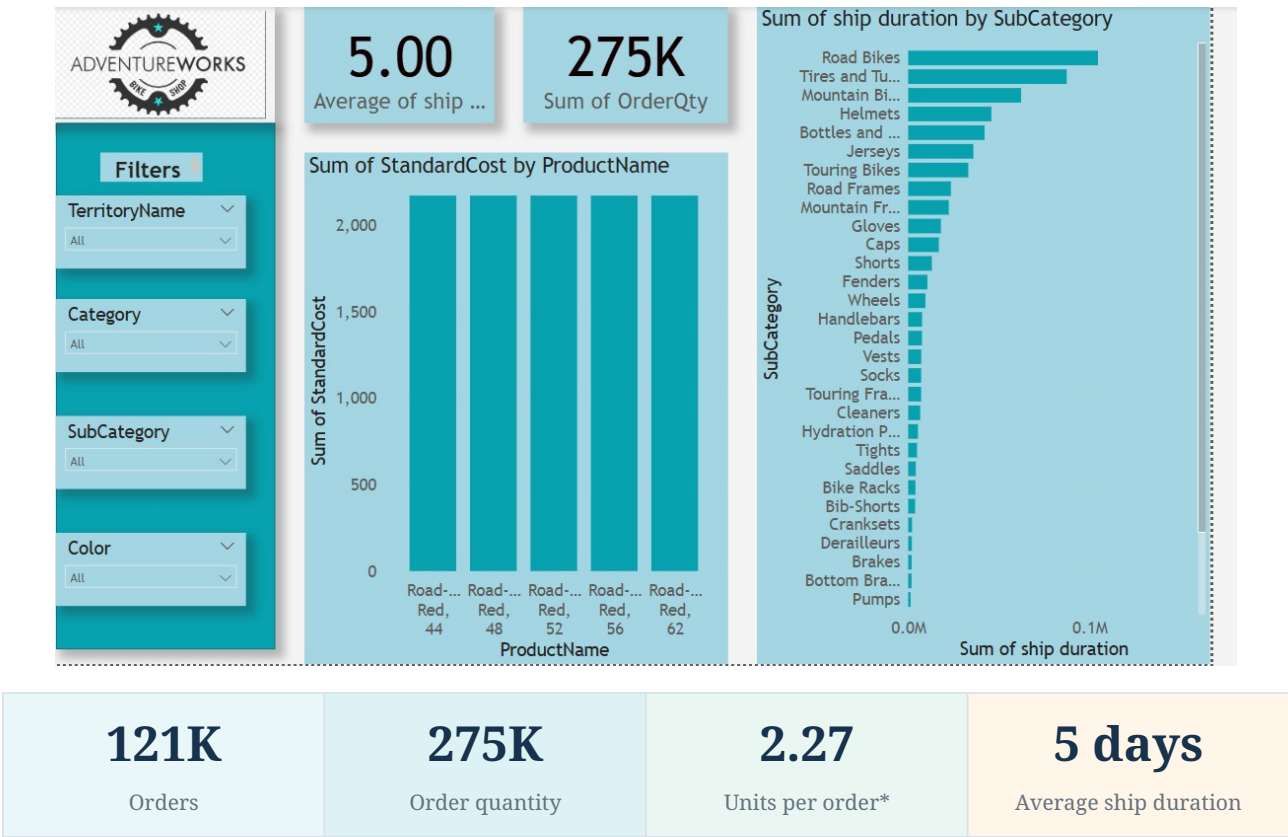
- A large NULL product category is present.
- Missing classification can distort category analysis.
- High-standard-cost products should be evaluated together with actual revenue and margin, not cost alone.

Data-quality control

Track the percentage of products with valid Category and Subcategory values. Missing master data should be treated as a reporting control issue, not only a visual formatting issue.

Operational scale is clear; delivery performance needs better KPI design

The shipping dashboard contains useful volume indicators but a cumulative shipping visual can be misleading.



*Approximate, calculated from rounded dashboard values: 275K / 121K.

Supported findings • Average shipping duration is displayed as 5 days. • Total order quantity is displayed as 275K units. • The business is operating at substantial transaction volume.	Measurement issue • The chart uses Sum of Ship Duration by SubCategory. • A subcategory with more orders can accumulate more shipping days even if individual deliveries are not slower. • The visual therefore should not be used to rank delivery speed.
Better operational KPIs Use Average and Median Shipping Days, % Orders Within SLA, Shipping Days by Territory, and late-order rate. These metrics turn the dashboard into an actionable service-performance tool.	

A dashboard is only as reliable as the definitions behind its cards

The KPI summary is valuable for executive monitoring, but definitions and denominators must be explicit.



KPI	Displayed / Derived	Portfolio interpretation
Revenue	\$109.85M	Strong scale; analyse mix and margin underneath.
Profit	\$9.37M	Positive profit, but efficiency depends on the margin denominator.
Profit %	9.33 displayed	Appears to equal Profit / Cost, not Profit / Revenue.
Avg. Order Value	\$905.45	Useful commercial KPI; segment by customer and territory.
Sold Products	266	Use with 504 total products to track catalogue utilisation.
Avg. Ship Duration	5 days	Add SLA compliance and median for operational control.

Dashboard design principle

Every KPI card should show an unambiguous business definition, a consistent denominator, and enough context to avoid false conclusions.

Five initiatives to improve the quality of growth

The recommendations are deliberately tied to evidence visible in the dashboards.

Priority	Initiative	Action	Expected management value
1	Protect profitability	Build revenue-cost-profit-margin views by product, category and territory.	Separate high revenue from truly high-margin performance.
2	Rationalise portfolio	Classify the 238 products with no recorded sales before delisting.	Reduce catalogue complexity without removing strategic SKUs blindly.
3	Diversify category mix	Cross-sell Components, Clothing and Accessories to Bike customers.	Reduce dependence on Bikes while using the strongest customer base.
4	Replicate regional success	Benchmark Southwest, Canada and Northwest against weaker territories.	Identify repeatable commercial drivers rather than guessing.
5	Upgrade operational KPIs	Replace cumulative shipping metrics with average, median and SLA measures.	Improve service management and logistics accountability.

90-day management roadmap

0-30 days	31-60 days	61-90 days
Validate KPI definitions; clean NULL classifications; tag unsold products.	Build margin views; segment unsold SKUs; benchmark top territories.	Launch targeted cross-sell tests; introduce shipping SLA scorecard; review portfolio actions.
Success condition The objective is not simply to sell more. It is to make products, customers, territories and orders contribute more effectively to sustainable profit.		

From dashboard builder to decision-support analyst

The value of the project is the translation of visuals into business questions, KPI validation and management actions.

Case-study conclusion

AdventureWorks generated \$109.85M in displayed revenue, but the analysis shows that strong top-line performance sits alongside category concentration, under-utilised catalogue capacity, limited margin visibility and KPI-definition risks. The recommended next step is a management reporting layer focused on profitability, portfolio productivity, cross-sell, regional benchmarking, operational service levels and data quality.

Portfolio positioning

Role: Business Intelligence / Data Analyst. I transformed sales dashboards into an executive decision-support case study, validated KPI logic, identified data-quality and aggregation issues, and translated evidence into an actionable business roadmap.

Skills demonstrated

Business analysis

- Problem framing and executive storytelling
- KPI definition and validation
- Commercial and profitability analysis
- Management recommendation design

BI / analytical thinking

- Power BI dashboard interpretation
- Metric quality and denominator checks
- Portfolio and category analysis
- Operational KPI redesign

Recommended next dashboard version

Add Revenue Growth %, Profit Margin %, Revenue per Customer, Repeat Purchase Rate, Revenue per SKU, Unsold Product %, Units Sold by Size, On-Time Shipping %, and Product Classification Completeness.

Claim Audit Register

All quantitative claims below were checked against the supplied dashboard screenshots or derived directly from those displayed values. No external data was used.

Load-bearing claim	Status	Decision / note
\$109.85M revenue and \$9.37M profit are displayed.	Verified	Retain as dashboard facts.
Bikes contribute about 86.1% of revenue.	Verified	Derived from \$94.65M / \$109.85M.
266 of 504 products recorded sales (52.8%).	Verified	Derived directly from displayed counts.
238 products show no recorded sales.	Verified	Displayed on cost/profitability dashboard and equals 504 - 266.
Conventional profit margin is about 8.53%.	Verified	Derived from \$9.37M / \$109.85M.
Displayed 9.33 is approximately Profit / Cost.	Verified	$\$9.37\text{M} / \$100.47\text{M} = \text{about } 9.33\%$.
2014 represents a true year-over-year decline.	Unsupported	Chart is lower, but completeness of 2014 is not established.
Road Bikes have the slowest shipping.	Unsupported	The visual uses SUM of shipping duration, not average or median.
Sizes 44 and 48 have the highest demand.	Unsupported	The visual sums list price by size, not units sold.
NULL product category indicates classification gaps.	Verified	A large NULL group is visible in the category chart.

Source & scope

Source: the supplied AdventureWorks Power BI dashboard screenshots. This portfolio report interprets only what the dashboards support; recommendations are analytical proposals for management review, not claims that the underlying business has already implemented them.

End of case study